

HOMEWORK 1

DISCRETE MATHEMATICS

MATH 208, FALL 2022

DUE: 08/31/2022

CHAPTER 1 - LOGICAL CONNECTIVES AND COMPOUND PROPOSITIONS

- Determine which of the following sentences are propositions. Assume you are speaking the sentence.
 - Today is Tuesday.
 - Why are you whining?
 - The Vikings are the worst team in professional sports.
 - This sentence has five words.
 - There is a black hole at the center of every galaxy.
- Construct truth tables for each of the following.
 - $\neg q \longrightarrow \neg p$.
 - $p \longrightarrow (q \wedge r)$. (You will need eight rows for this one.)
- Perform the indicated bit string operations. The bit strings are given in groups of four bits each for ease of reading.
 - $(1001\ 0101 \oplus 1010\ 0110) \wedge 1100\ 1000$
 - $(1110\ 1010 \wedge 0101\ 0010) \vee 0111\ 1001$
 - $(1111\ 0011 \vee 0111\ 0101) \wedge (0010\ 0010 \vee 0110\ 0100)$
- Let s be the proposition *It is snowing* and f be the proposition *It is below freezing*. Convert the following English sentences into statements using the symbols s , f and logical connectives.
 - It is snowing and it is below freezing.
 - If it is snowing, then it is below freezing.
- Like any natural language, spoken English can be ambiguous. It is sometimes necessary to rely on context, voice inflection or other clues, to correctly interpret a sentence. Propositional logic, correctly written, is never ambiguous. Express the following two sentence pairs in symbolic form that correctly conveys the intended meaning.
 - Dinner comes with peas and carrots or french fries.*
Intended: You get either the peas/carrots combination, or else you get french fries.
Intended: You get peas together with your choice of one of carrots or french fries.
 - A similar scenario with algebraic operations: *Two plus three times four.*
Intended: add two and three, and multiply the total by four.
Intended: add two to the product of three and four.

CHAPTER 2 - LOGICAL EQUIVALENCE

1. Use a truth table to show $p \rightarrow q \equiv \neg p \vee q$.
2. Use a truth table to show $(p \wedge q) \rightarrow r \equiv p \rightarrow (q \rightarrow r)$.
3. Use a truth table to show $p \rightarrow q \equiv \neg q \rightarrow \neg p$.
4. Consider the implication *If I work, then I get paid*.
 - a) Write the converse of the implication.
 - b) Write the inverse of the implication.
 - c) Write the contrapositive of the implication.
 - d) Write the inverse of the contrapositive of the implication.
5. Consider your answers to Problem 4. Which of the four items are logically equivalent to the implication? Which of the four are not logically equivalent to the implication but are logically equivalent to each other?
6. Use a truth table to show $[p \rightarrow (q \rightarrow r)] \not\equiv [(p \rightarrow q) \rightarrow r]$.
7. Use a truth table to show that $(p \wedge q) \rightarrow p$ is a tautology.
8. Give a proof of $\neg p \rightarrow (p \rightarrow q) \equiv \top$ using the Fundamental Logical Equivalences.